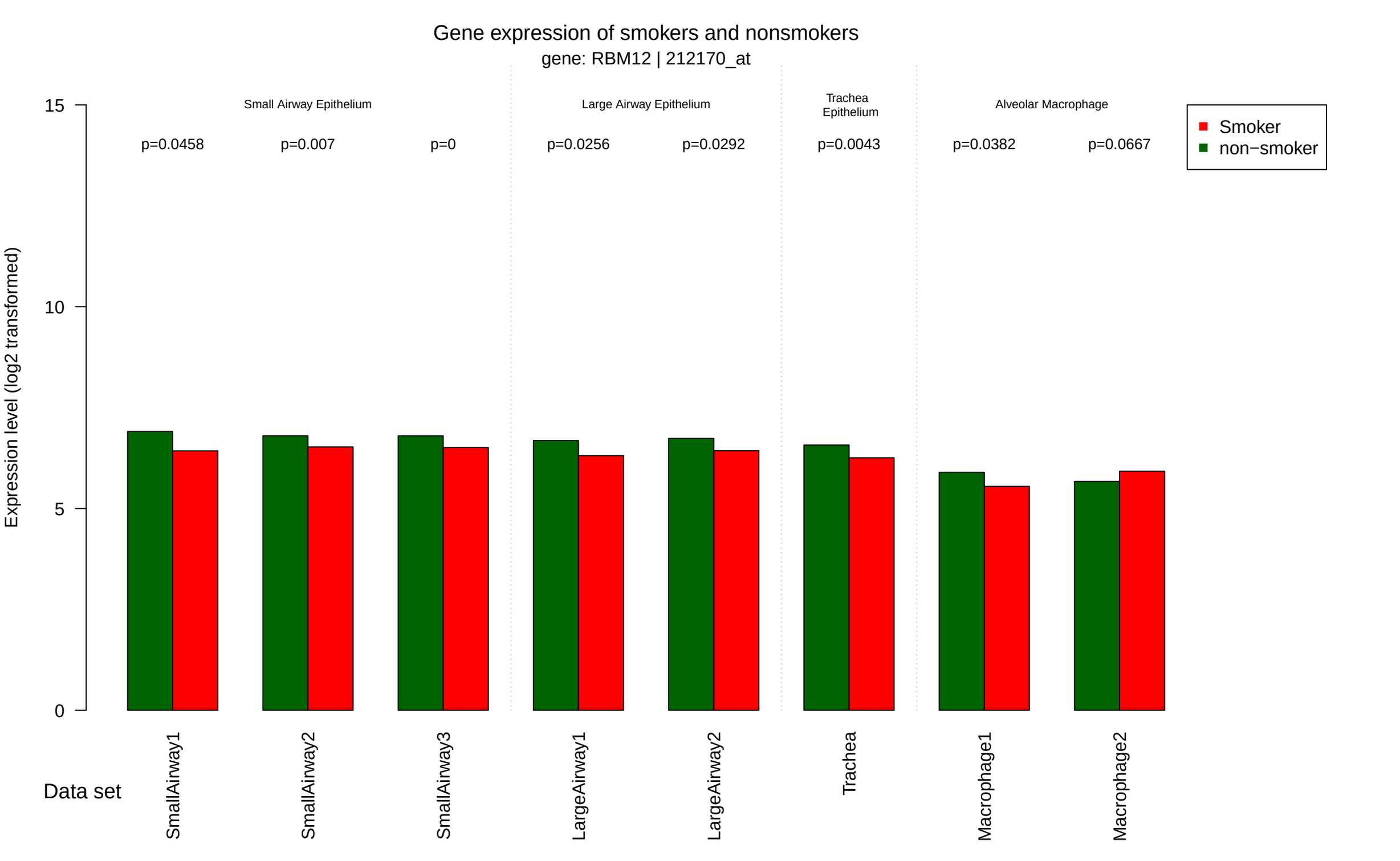
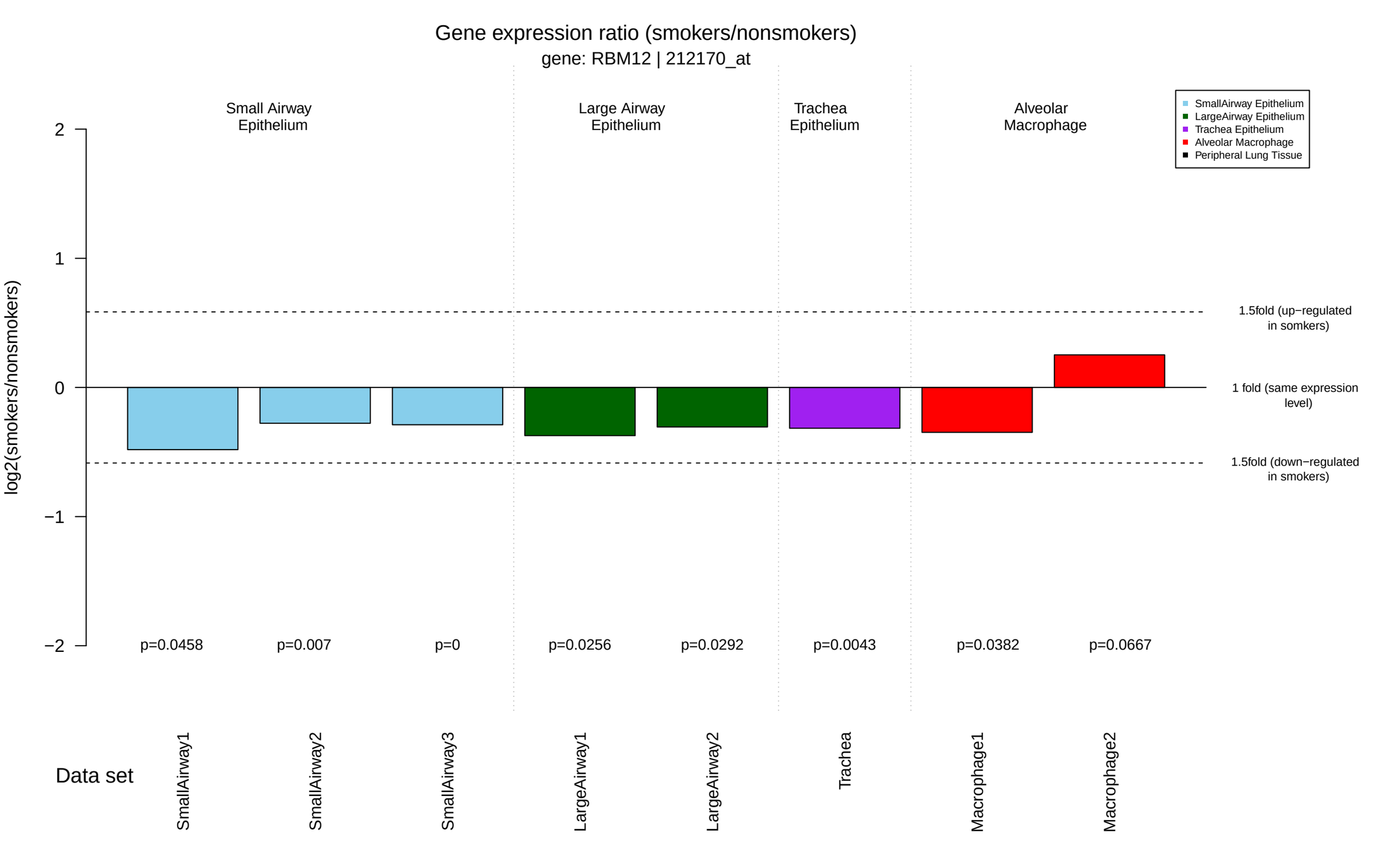
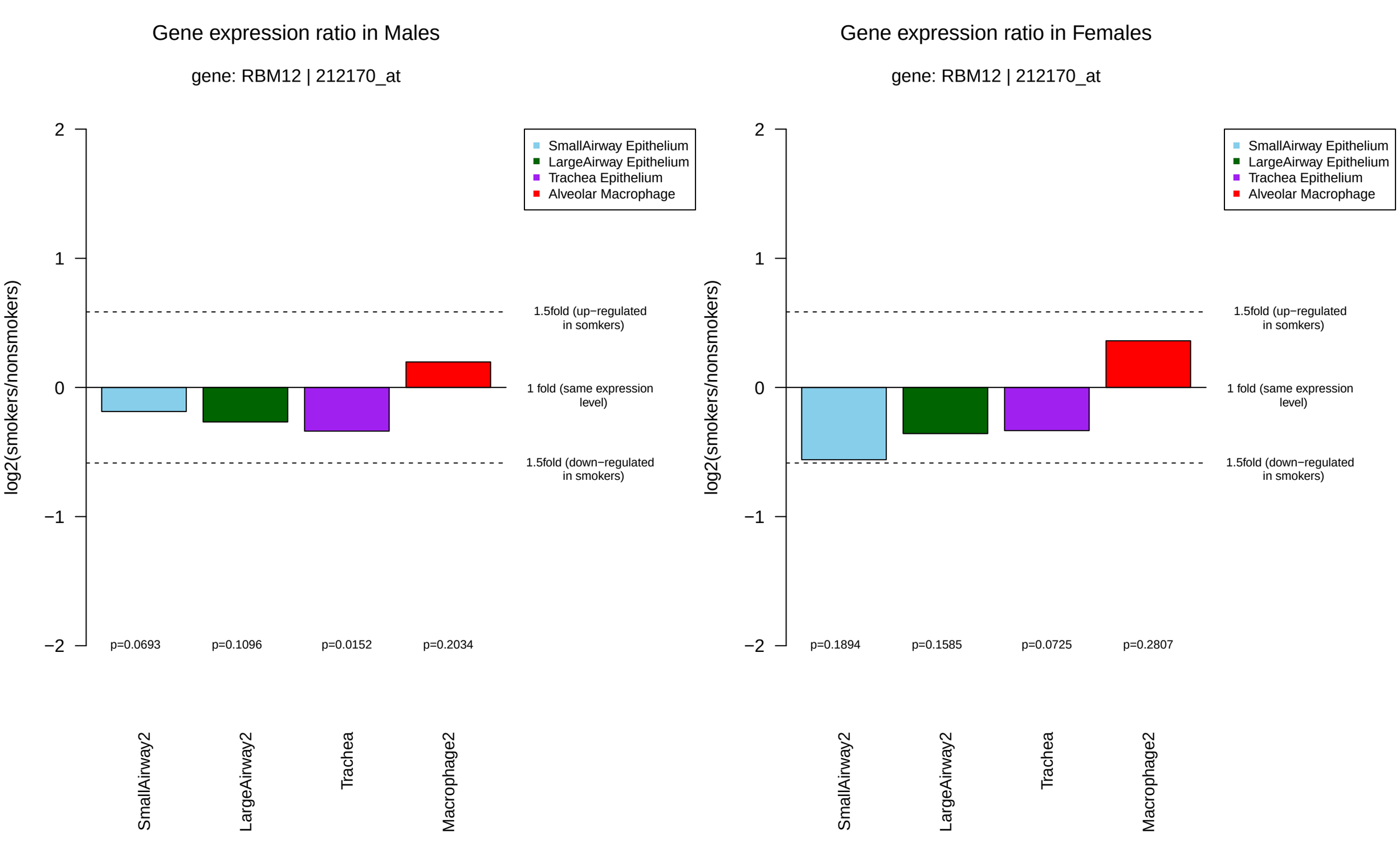
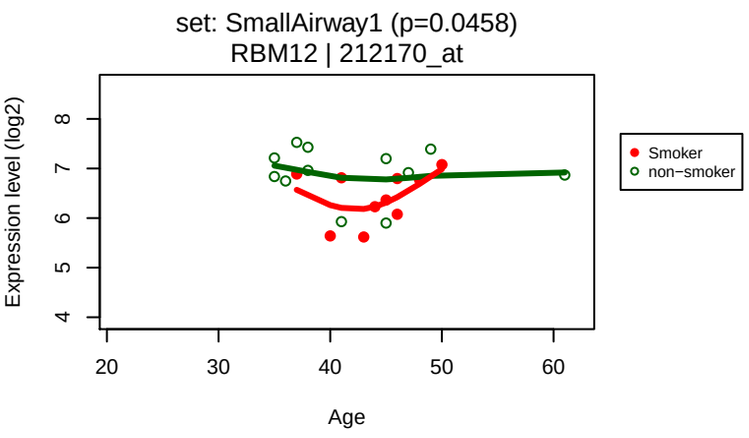
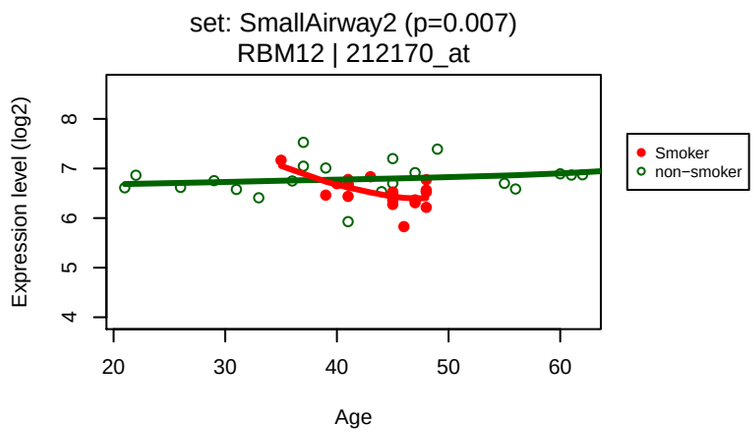
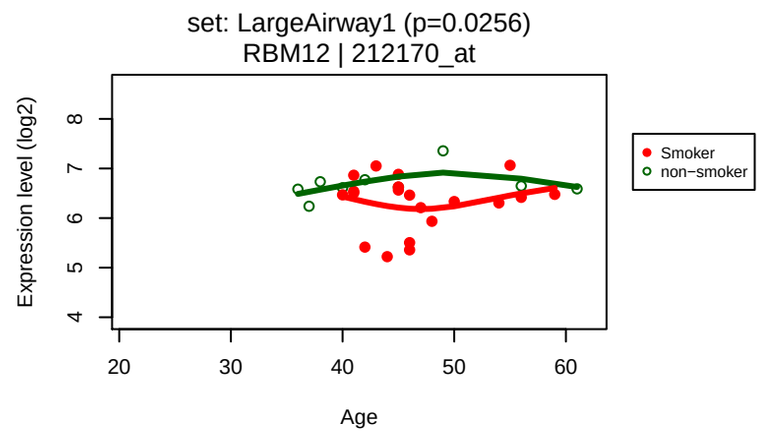

Gene expression of smokers and nonsmokers
gene: RBM12 | 212170_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0458
p=0.007
p=0
p=0.0256
p=0.0292
p=0.0043
p=0.0382
p=0.0667
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

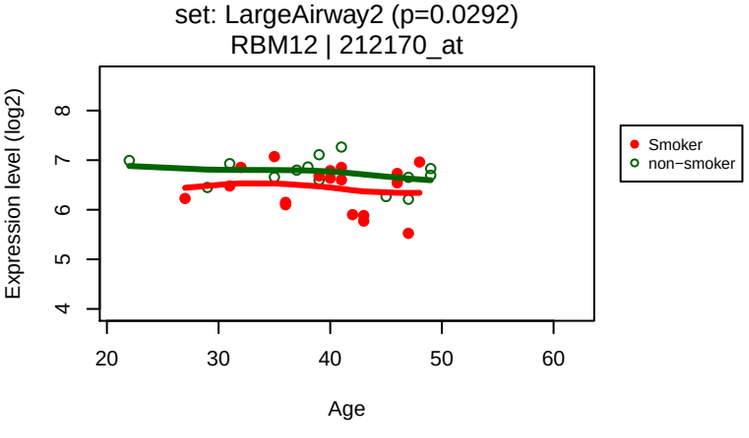

Gene expression ratio (smokers/nonsmokers)
gene: RBM12 | 212170_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0458
p=0.007
p=0
p=0.0256
p=0.0292
p=0.0043
p=0.0382
p=0.0667
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

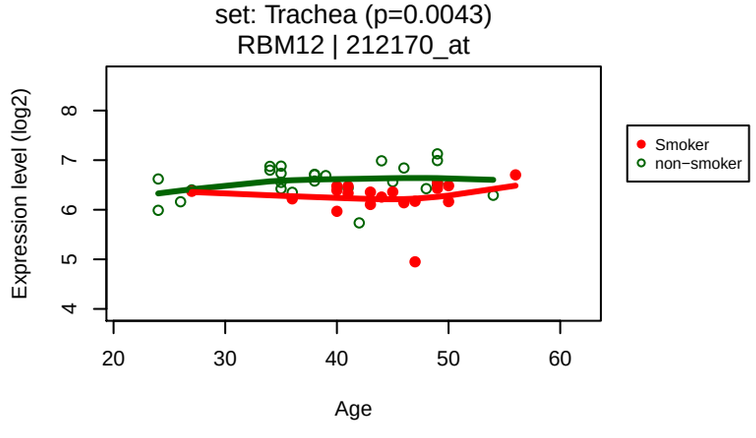

Gene expression ratio in Males
gene: RBM12 | 212170_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0693
p=0.1096
p=0.0152
p=0.2034
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RBM12 | 212170_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1894
p=0.1585
p=0.0725
p=0.2807
SmallAirway2
LargeAirway2
Trachea
Macrophage2

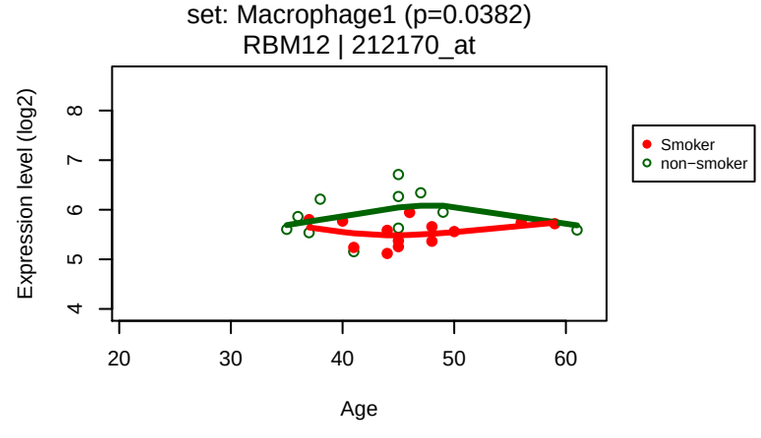

set: SmallAirway1 (p=0.0458)
RBM12 | 212170_at
Expression level (log2)
Age
Smoker
non−smoker

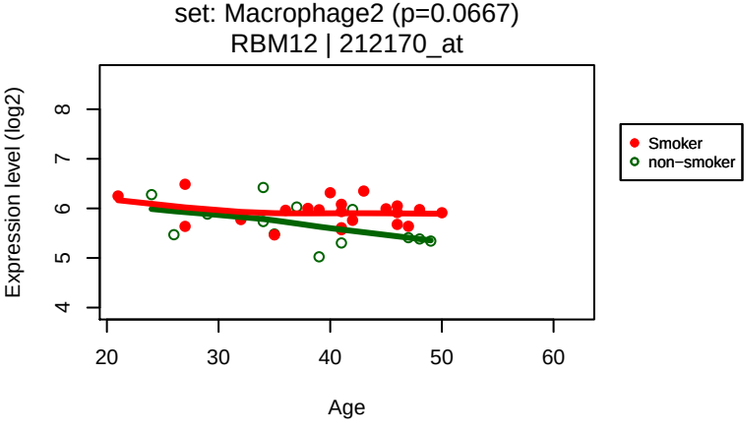

set: SmallAirway2 (p=0.007)
RBM12 | 212170_at
Expression level (log2)
Age
Smoker
non−smoker

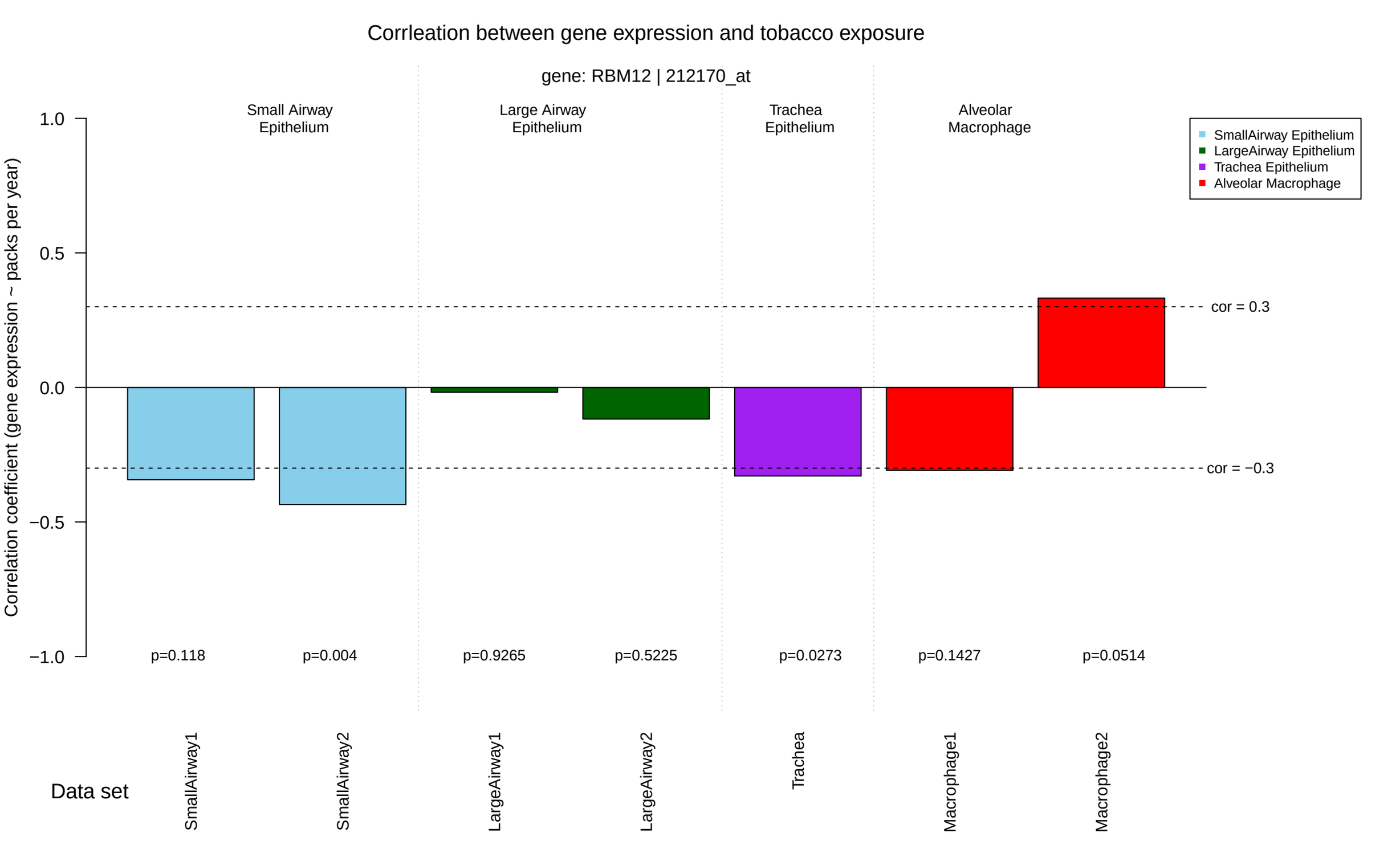

set: LargeAirway1 (p=0.0256)
RBM12 | 212170_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0292)
RBM12 | 212170_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0043)
RBM12 | 212170_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0382)
RBM12 | 212170_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0667)
RBM12 | 212170_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RBM12 | 212170_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.118
p=0.004
p=0.9265
p=0.5225
p=0.0273
p=0.1427
p=0.0514
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2